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Iterative calibration of medical digital twins via adaptive estimators

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While iterative calibration of computational models is a fundamental aspect of digital twins, it has been largely overlooked. Instead of focusing on parameter identification for static models, the implementation of digital twins requires not only high-resolution computational models but also the ability to assimilate patient-specific data continuously. Here, we review adaptive estimator algorithms and how this can address iterative calibration of digital twins, enabling the estimation of unmeasurable states while continuously adapting model parameters. Integrating adaptive estimators into digital twins offers a paradigm shift: transforming them from static representations into living, evolving systems that advance personalized medicine.

KEYWORDS

adaptive estimators, filters, medical digital twins, observers, personalized medicine

1 Introduction

Since their conceptualization in the early 2000s, Digital Twins (DTs) have been built on three fundamental pillars: (i) a virtual model, (ii) a physical counterpart, and (iii) dynamic updating mechanisms that synchronize the first two pillars through continuous data exchange [1–3]. The concept of digital twins was initially implemented for physical industrial products and services [4]. Nowadays, the aforementioned approach is widely applied in many sectors that assist forecasts and minimize emergency behavior by modeling complex systems under a wide range of real-world conditions [3, 5]. For instance, NASA created virtual replicas of spacecraft for design, maintenance, and monitoring [6]. Some companies employ digital twins to enhance energy production [7] or diagnose and anticipate real-world issues in their facilities [8]. Additionally, local and state governments are employing digital twins technology for urban planning and disaster management [9, 10].

Developing digital twins from scratch is a complex task that requires precise modeling and the integration of various data sources. To address these challenges, both industry and academia are focusing on creating comprehensive digital twin platforms. These platforms serve as repositories of ready-to-use or reusable models, data, functions, and tool assets, significantly increasing accessibility for organizations looking to implement digital twin technology on a large-scale and efficient manner [11, 12]. Consequently, numerous digital twin platforms have been proposed in the literature [13], with some aiming to be provided as open-source software [14].

In the medical field, Medical Digital Twins (MDTs) have been envisioned as patient-specific computational models capable of supporting diagnosis, prognosis, and therapy design [15–17]. MDTs offer promising solutions for testing drug efficacy, optimizing

treatment plans, and predicting disease progression [17–19]. Some recent applications include modeling Tuberculosis infection [20], modeling virtual patient cohorts during viral infections [21], creating immune system simulation platforms [22, 23], developing pneumonia MDTs to support decision-making processes in intensive care units [24], developing breast cancer MDTs [25], and developing MDTs for leukemia [19].

The workflow for developing MDTs, as illustrated in Figure 1, involves utilizing disease-relevant biological data to establish a generic computational model. This model is then validated using both population and patient-specific data, resulting in a personalized model. This personalized model can then be applied to the prediction and simulation of diseases [19].

1.1 Bottlenecks in calibrating medical digital twins

The boundaries that govern current MDT applications are particularly evident in their dependence on static mechanistic models. These models are typically parameterized once at the beginning of the model identification process using a predefined dataset. However, these computational models remain *static* for a fixed period of time until a new and sufficiently large dataset becomes available, at which point the model parameter identification procedure is triggered once more. Therefore, while beneficial in providing a foundational understanding of complex systems, current digital twins often fail to keep up with ever-evolving data. Digital twins require continuous model updating that evolves at the same pace as new data becomes available, in order to reflect their physical counterparts accurately.

As stated in Wright and Davidson [26], the ability to dynamically update or adjust the model as new data becomes available is a crucial component that differentiates a digital twin from a conventional mathematical model. Traditional mathematical and statistical models train on available data to produce a fitted curve that can be used for predicting some outcome, but with the addition of new data, the majority of these models need to be refitted on the whole data set. There are a few nuanced exceptions to this, with recursive least squares regression (RLS), which updates model parameters upon the addition of a single data point without refitting against the entire data set, being one such example [27]. However, despite having the ability to update model parameters without refitting the whole model when new data is added, there comes a point when RLS can no longer reliably update model parameters without being refit to the entire dataset again [27]. In essence, if a model needs to be refit to the data at any point in time, it is not an adaptive model, and thus, not a digital twin.

Digital twins applications need to consider data assimilation frameworks that allow for continuous updates and refinements of the mechanistic model. This means going from *static* to *self adaptive* systems, which are capable of responding to real-time scenarios. In control theory, an observer is an algorithm that provides an estimate of the state variables of a real system by combining information from system inputs, system outputs, and a model of the system's dynamics [28–30]. Adaptive estimators, on the other hand, offer the ability to estimate unmeasured states while enabling real-time parameter recalibration [31]. Adaptive observers differ from traditional observer designs, such as the Luenberger observer or Kalman filter, by extending the observer paradigm to joint state-parameter estimation. While traditional observers assume that

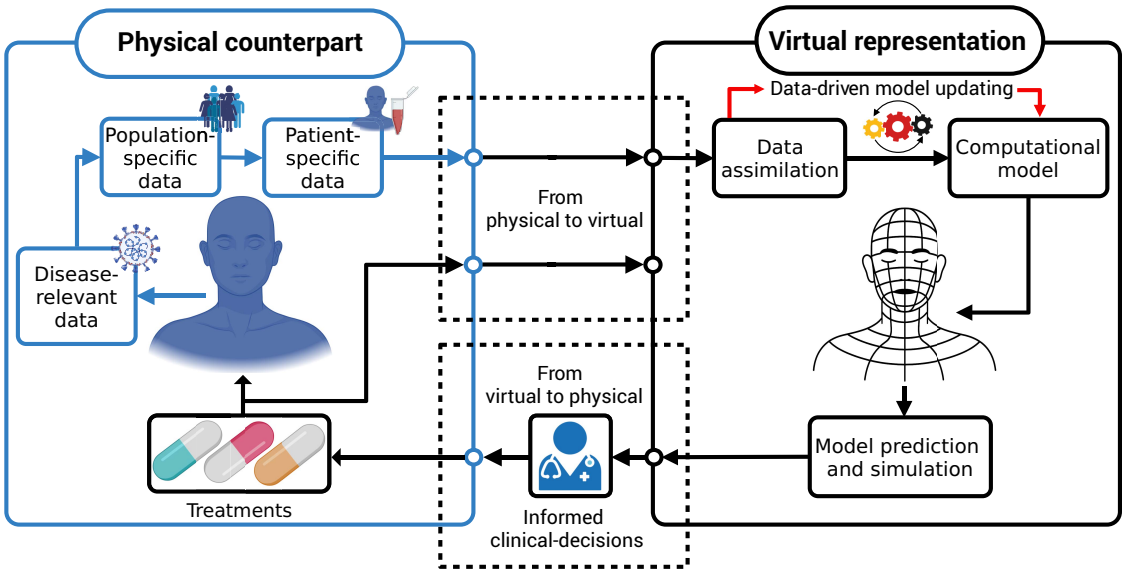

FIGURE 1
Medical digital twin framework. The essential parts for the development of a digital twin, according to the National Academies of Sciences, Engineering and Medicine. A crucial part is that information flows bidirectionally between the physical and virtual counterparts. The human to digital direction allows the recalibrating of the model to make accurate predictions. This allows valuable information for the decision-making process of practitioners to become available. Created in BioRender. Sereno, J. (2025) BioRender.com/zt4sx9x.

parameters are fixed and known [32], adaptive observers integrate an online parameter adaptation law into their dynamics [33].

Although there are proposals in the literature for extended versions of the Luenberger oberver or Kalman filter aimed at jointly estimating states and parameters [34, 35], these proposals often lack certain desired features, such as the ability to explicitly track unknown parameters [36]. This limitation makes the adaptive observer design more suitable for digital twins. Integrating mechanistic computational models with adaptive observer algorithms would enable the transition from *static* computational systems to digital twins.

In this review, we present a framework for calibrating digital twins using adaptive observers. The next section focuses on classical observers and their limitations. Then, fundamental concepts for adaptive observers, stochastic filters, and the adaptive digital twin framework are presented. Next, we discuss the applicability of digital twins to viral infections, drug-resistant pathogens, and co-infections. We end with conclusions and relevant insights.

2 Observers and estimators

Reconstructing unmeasured states from partial measurable outputs of a system is a central problem in control theory [30]. Classical observers, like the Luenberger case, ensure asymptotic convergence of the estimated states to their actual values through the design process. It is worth mentioning that an estimator is a more general concept—any algorithm or procedure that produces an estimate of an unknown quantity. This could be a state, a parameter, or even a noise signal. For example, the Kalman filter case is based on a probabilistic estimator that minimizes the mean square estimation error by incorporating both the process and measurement noise statistics [29].

While the Kalman filter was initially developed for linear systems, the Extended Kalman Filter or Extended Kalman Observer extends these concepts to nonlinear systems by linearizing the state dynamics at each time step [30]. In general, the Luenberger observer has a simple implementation and low computational cost, whereas the Kalman filter is more computationally demanding, but offers good performance under noise and uncertainty. The above-mentioned observers typically assume full knowledge of system parameters. When this assumption is not satisfied, their performance can degrade significantly [32].

2.1 Adaptive observers

The idea of combining state observation with parameter adaptation dates back to early works, such as Luders and Narendra [37] and Kreisselmeier [33], where convergence under restrictive conditions was established. Later, in Bastin and Gevers [38], a systematic design approach for single-output nonlinear systems was introduced, employing canonical transformations to simplify the parameter dependence. With these foundational results, works such as Besancon et al. [31] and Zhang [36] extended the framework to time-varying systems and state-affine nonlinear systems, overcoming many of the structural limitations of the earlier approaches. Formally, adaptive observers consider state-affine systems of the form:

$$\dot{x}(t) = A(t)x(t) + B(t)u(t) + \Psi(t)\theta \tag{1a}$$

$$y(t) = C(t)x(t), \tag{1b}$$

where $x(t) \in \mathbb{R}^{n_x}$, $u(t) \in \mathbb{R}^{n_u}$, and $y(t) \in \mathbb{R}^m$ are the states, inputs, and outputs of the system, respectively; matrices $A(t)$, $B(t)$, and $C(t)$ are known time varying matrices; $\theta \in \mathbb{R}^p$ is the unknown vector of parameters that is assumed constant; and $\Psi(t) \in \mathbb{R}^{n_x \times p}$ is a matrix of known signals. Unlike the discussed traditional observers, the adaptive observer design takes advantage of the state-affine form with respect to θ and constructs a coupled scheme. This results in a stabilizing state observer, coupled with a parameter adaptation law [36]. The convergence of adaptive observers rests on two key assumptions: (i) the existence of a stabilizing observer gain $K(t)$ for the nominal system and (ii) persistent excitation (PE) of the regressor signals, which ensures identifiability of the parameters. Under these conditions, adaptive observers achieve global exponential convergence in noise-free settings and robustness in the presence of bounded disturbances.

Adaptive observers differentiate from the classical observers in different aspects as presented in Table 1. First, it is more suitable for both state and parameter estimation. Second, it has achievable design requirements. Third, it has global and exponential convergence when noise is not considered and bounded mean when noise is considered. Fourth, it is robust against bounded disturbances and parameters. Lastly, it has a moderate-to-high computational complexity, depending on the problem dimension, i.e., the considered number of states-parameters to be estimated. Table 1 summarizes the main applications for each type of observer.

Although the adaptive observer was originally developed for joint state-parameter estimation, it is not the sole proposal available in the literature for addressing this task. For instance, in Afri et al. [34], the authors present the theoretical formulation of a nonlinear Luenberger observer approach for state and parameter estimation. This proposal offers a simpler, globally stable alternative to the adaptive observers; however, in this case, parameter estimates are not directly produced, limiting its role to cases in which tracking parameters is not the primary task. In Deng et al. [35] and Van Der Merwe R, Wan et al. [39], extended formulations of the Unscented Kalman Filter (UKF) can be found for joint state-parameter estimation; in these approaches, the augmented state method is implemented.

The aforementioned approaches come with high computational costs, and, moreover, the augmented state methodology increases system dimensions and leads to tuning issues that could result in biased parameter estimates or even filter divergence. For a comprehensive discussion on Gaussian filters used in state and parameter estimation, readers may refer to Afshari et al. [40]. Joint state-parameter estimation algorithms are summarized in Table 2.

The adaptive observer, while successfully applied in real settings [41], has two notable drawbacks: (i) its computational demand—this method necessitates the calculation of matrix inverses, and the computational cost escalates as the problem dimension increases; and (ii) its reliance on a continuous form

TABLE 1 Comparison between Luenberger, Kalman, and Adaptive observers.

Aspect	Luenberger observer	Kalman filter	Adaptive observer
System class	Deterministic	Stochastic	LTV, MIMO, and state-affine
Estimation objective	States only	States (optimal in mean-square sense)	States + parameters
Design requirement	Observability of (A, C)	Uniform complete observability	Stabilizability of (A, C) + persistent excitation
Convergence guarantee	Asymptotic (if detectable)	Under strong observability + noise assumptions	Global exponential (noise free); bounded mean under noise
Robustness	Weak to parametric mismatch	Robust to Gaussian noise	Robust to bounded disturbances and parameter drift
Computational complexity	Low (static gain)	Moderate/high (covariance)	Moderate/high (observer + adaptation law)
Applications	Classical control	Tracking, navigation	Adaptive control, fault detection, digital twins

TABLE 2 Summary of joint state-parameter approaches.

Observer technique	References	Key findings
Luenberger-like observers	[34, 85]	Proof of existence for linear systems with unknown parameters and robustness for output noise
Kalman-like observers	[35, 39, 40, 86]	Ensemble Kalman filters estimate parameter-states in high dimensions and nonlinearities
Adaptive observers	[31, 36, 87, 88]	Its transformation allows decoupling estimation of states and parameters with exponential convergence

formulation—this limitation becomes evident when implementing this strategy in situations where sparse sensor measurements are prevalent. These challenges highlight the need for alternative approaches that can maintain efficiency and effectiveness in environments with limited data. Exploring new methods that reduce computational complexity or adapt to discrete measurement scenarios could enhance the applicability of observer designs in diverse real-world contexts.

Figure 2 describes our proposal of adaptive observers for digital twins, i.e., the adaptive digital twin framework. In this case, adaptive observer algorithms are used for dynamical updating of model parameters (see the parameter-updating block) at the same pace that new data becomes available and, more importantly, without triggering complete retraining cycles of complex optimization methods for parameter identification—*Static vs Continuous*. This approach has the potential to enable better monitoring of key biological markers, as well as enhance predictions for virtual drug and therapy testing. This would enable a new paradigm in precision medicine.

2.2 Stochastic adaptive filters

Consider the following stochastic linear differential equation with an unknown vector parameter $\theta(t)$ [42]:

$$dx(t) = (A_0(\theta, t) + A_1(\theta, t)x(t))dt + b(t)dW_1(t), \tag{2}$$

with the observation process

$$dy(t) = (C_0(t) + C_1(t)x(t))dt + B(t)dW_2 \tag{3}$$

where $x(t_0) = x_0$, $x(t) \in \mathbb{R}^n$ is the state vector, $y(t) \in \mathbb{R}^n$ is the linear observation vector such that the observation matrix $A(t)$ is invertible, $\theta(t) \in \mathbb{R}^p$, $p \leq n \times n + n$ is a vector of unknown parameters, $W_1(t)$ and $W_2(t)$ are Wiener processes independent of each other and of the initial condition x_0. The estimation problem consists in finding the best estimates for the state vector $x(t)$ and unknown parameters $\theta(t)$ by using the observation measurements $y(t)$.

One of the approaches that estimates the state and parameters is presented in Yaakov [43], where the maximum likelihood estimation method is employed to simultaneously estimate the state and parameters for linear discrete-time dynamic systems driven by additive white Gaussian noises. The parameters to be estimated are considered unknown and constants. The whole problem is separated into two interconnected linear problems: one for the state, and the other one for the parameters. Employing the extended Kalman filter (EKF) and the expectation-maximization (EM) algorithms. Togneri and Deng [44] reported the problem of joint state and parameter estimation for stochastic hidden dynamic models with switching states. The idea is to define a new state vector that is composed of the parameters and the state vector, and then conduct several experiments where the EKF or EM algorithms are employed.

Another approach that jointly identifies the state vector and parameters is proposed in Basin et al. [42] for linear stochastic systems. To cope with the reconstruction of state and parameter identification, the unknown parameters $\theta(t)$ are modeled as Wiener processes, and then a new state vector that is composed of the state vector and the unknown parameters is defined, $z(t) = [x^\top(t)\ \theta^\top(t)]^\top$. Finally, a new set of stochastic differential equations with a polynomial form is obtained and the optimal estimate for the new state vector is computed. Some applications of parameter identification are found in Nguyen et al. [45], where the least square method is employed to identify the parameters of a six-DOF manipulator.

Sun and Yang [46] employed the Kalman filter with the Maximum Likelihood (KF-ML) method to handle the joint

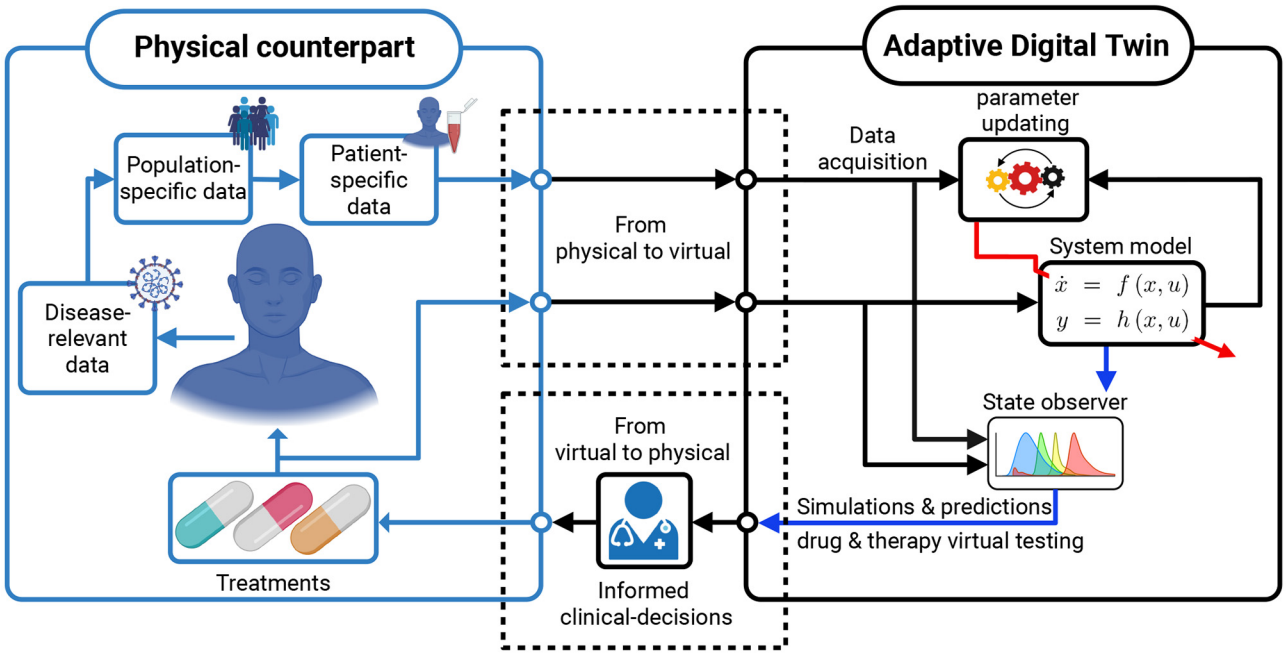

FIGURE 2
Adaptive digital twin framework. The dynamical update of the virtual representation is performed through an adaptive observer algorithm that perform, in a continuous manner, the estimation of unmeasurable states (virtual sensors, state observer block) and the model parameter recalibration (dynamical model update from physical twin data, parameter updating block). Created in BioRender. Sereno, J. (2025) BioRender.com/up1avn5.

parameter identification and state estimation problem for a fault-tolerant space robot system. The problem is divided into two stages: the first one employs the Kalman filter algorithm to obtain an estimation of the states that is dependent on the unknown parameter; and the second stage employs the ML technique to estimate the unknown parameter. Finally, the estimated parameter is substituted into the parameterized state estimation to obtain the desired solution. For a review of different methods of adaptive filters, see [40].

3 Developing medical digital twins for infectious diseases

The immune system plays a critical role in the development and progression of numerous significant diseases [17]. Consequently, the development of a reliable software representation of the immune system, known as an Immune Digital Twin (IDT), could significantly impact healthcare and biological research [47]. As expected, there is a large interdisciplinary and international research community focused on developing an IDT [16, 48, 49]. Yet the immune system response is intricate and diverse for various medical conditions and individuals [50], which presents significant technical challenges that require careful consideration.

Accordingly, the integration of adaptive observers into an IDT represents a paradigm shift from *static* to *self adaptive* computational models that will help practitioners toward therapy selection, optimal decision making for specific patients, and allow disease prediction.

Currently, there are promising applications of MDTs, such as the living heart project [51, 52] or the artificial pancreas project [53, 54]. This adaptive digital twin framework is expected to broaden these promising applications by enabling systems capable of continuously updating and iteratively reparameterizing, closing the gap between static models and true digital twins. The following subsections demonstrate how adaptive observers can revolutionize personalized medicine across different infectious disease contexts.

3.1 Viral infections

Viral infections pose unique challenges for immune DT implementation due to their rapid mutations and complex interactions with the host immune response. Mathematical models of viral infections have been instrumental in understanding disease progression, but often lack the adaptability required for personalized treatment strategies [55–57]. Monitoring immunological markers can be expensive, and some tests for certain markers are not feasible to perform. The use of adaptive observers can enable the monitoring of immune system responses by using blood samples [58, 59].

3.1.1 HIV infection

For HIV infection, where long-term management and treatment strategies are critical, adaptive observers can predict CD4+ T cell counts, viral reservoir dynamics, and drug resistance evolution [60]. This tracking procedure can be performed by the adaptive observer using patient-specific data pertinent to the

HIV infection, such as viral load data and CD4+ T cell counts. Adaptive observers require a dynamical mechanistic model of the HIV infection, in which case it would be possible to estimate and monitor changes in infected cells and reservoirs [61]. These capabilities play an important role for aging HIV patients, where immunological heterogeneity and treatment complications require personalized therapeutic approaches [62]. Medical immune digital twins are pivotal for the evaluation of the long-term outcomes of purging strategies and their effect on the HIV reservoir, which are still largely fragmented [60].

3.1.2 Influenza infection

In influenza infections, adaptive observers can monitor the three stages of the infection process, which are initial viral replication, immune response activation, and either resolution of infection or further complications thereof [63, 64]. In a similar manner, the adaptive framework, coupled with a relevant influenza model and the pathogen load measurements, enables prediction of disease progression, thus allowing for optimal timing of antiviral interventions, which are particularly crucial in elderly patients where immune responses are often impaired [65]. Moreover, recent applications have demonstrated the utility of mathematical models in predicting influenza-specific CD8+ T cell responses, inflammatory responses, and optimizing treatment strategies [59, 66]. Thus, enabling the monitoring of inflammatory responses with adaptive observers could improve diagnosis and prognosis.

3.2 Antibiotic resistance

The emergence of drug-resistant pathogens represents one of the most pressing challenges in modern medicine, with traditional static models inadequately capturing the dynamic evolution of resistance mechanisms [67, 68]. Adaptive observers offer a transformative approach by enabling continuous tracking of pathogen evolution and real-time adjustment of therapeutic strategies.

In the context of antibiotic resistance, adaptive observers can monitor bacterial population dynamics, resistance gene expression, and the emergence of resistant subpopulations during treatment. This capability is essential for implementing collateral sensitivity-based cycling therapies, where the order and timing of antibiotic administration must be carefully optimized to prevent resistance development [69, 70].

Computational approaches have shown promise in developing cutting-edge methods to personalize treatment strategies and mitigate drug resistance evolution [69]. In this scenario, by iteratively updating model parameters based on patient-specific measurements, adaptive observers can predict when current therapeutic approaches may fail and recommend alternative treatment strategies before clinical deterioration occurs [71]. This proactive approach represents a significant advancement over reactive treatment modifications based on observed therapeutic failure. Modeling heterogeneous bacterial populations and predicting the evolution of resistance under different selective pressures provides clinicians with unprecedented insight into treatment optimization [69, 72, 73].

3.3 Co-infections

Co-infections, particularly viral-bacterial combinations such as influenza and *Streptococcus pneumoniae*, represent some of the most complex scenarios for IDT implementation. The synergistic interactions between pathogens, combined with host immune system modulation, create highly dynamic systems that require sophisticated adaptive control strategies [74, 75].

Alveolar macrophages serve as critical sentinels in respiratory co-infections, with their functional states determining infection outcomes through complex polarization mechanisms between M1 (inflammatory) and M2 (healing) phenotypes [76, 77]. Adaptive observers can track these macrophage dynamics, monitoring key cytokine profiles, such as IFN-γ and IL-6, that determine susceptibility to secondary bacterial infections [78, 79]. This capability of the adaptive framework addresses a major knowledge gap in co-infection modeling, where static representations fail to capture the spatial and temporal modulation of immune cell functions. The adaptive framework enables integration of multi-scale data, from cellular immune responses to tissue-level pathogen dynamics. For influenza-*pneumococcal* co-infections, adaptive observers can track the transition from viral-mediated immune suppression to bacterial invasion and systemic dissemination [63, 80]. This multi-pathogen adaptive modeling approach provides insights into the timing-dependent effects of therapeutic interventions, enabling optimization of both antiviral and antibiotic therapies.

It is important to mention that most digital twins remain in their early stages and are often far from clinical practice. Up to the author's knowledge, limited medical DT-based systems have been validated in routine clinical practice. A successful example is the artificial pancreas [81]. Heart digital twins have achieved notable advancements, such as the patient-specific cardiac digital twins for ventricular tachycardia ablation guidance [82]. However, these two examples uses are still limited to research settings as a proof-of-concept [83]. Most current MDT applications primarily perform forecasting or retrospective patient-specific simulations, which limit the possibility of a direct, prospective, head-to-head comparison between MDT models. Lastly, MDT proposal reliability and robustness are crucial. Modern MDT pipelines increasingly incorporate uncertainty quantification techniques to explicitly characterize model and data uncertainty and provide more trustworthy, clinically oriented medical digital twins [84].

In the context of adaptive observers, robustness to uncertainty is typically achieved through structural properties of the estimation law, such as input-to-state stability and Lyapunov-based adaptation mechanisms. These features allow the observer to remain stable and provide bounded estimation errors when the underlying model parameters, measurements, or disturbance inputs are uncertain bounded open sets. Additionally, adaptive observer frameworks can be combined with stochastic or set-based uncertainty representations, enabling the estimation algorithm to explicitly account for parameter uncertainties through bounded parameter estimation. Overall, these robustness properties make adaptive observers suitable for medical digital twin implementations, where data quality and physiological variability are inherent sources of uncertainty.

4 Conclusion

Medical Digital Twins hold enormous promise for personalized medicine, but their current reliance on static mechanistic models limits their effectiveness. Digital twins must continuously evolve, assimilating patient-specific data in real time to remain faithful to the biological system they represent. In this work, we propose that adaptive observers provide a natural and powerful framework to achieve this transformation. Unlike classical observers, such as the Luenberger or Kalman filter, adaptive observers jointly estimate unmeasured states and unknown parameters, enabling continuous recalibration of the computational model.

By bridging control theory with biomedical applications, adaptive observers open the door to Medical Digital Twins that are no longer static approximations but are instead dynamic, living systems. Such twins could adapt to disease progression, account for inter- and intra-patient variability, and anticipate treatment resistance. The cases of viral infections, drug-resistant pathogens, and co-infections illustrate the urgency of this capability, where evolving dynamics demand equally adaptive models.

Looking ahead, integrating adaptive observers into MDTs represents more than a technical improvement: it is a paradigm shift. It moves digital twins from being descriptive tools to prescriptive, evolving companions in clinical decision-making. The potential reward is transformative: adaptive observer-driven digital twins could become the cornerstone of a new era in precision medicine.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

JS: Conceptualization, Writing – review & editing, Writing – original draft. HN: Writing – review & editing, Conceptualization, Writing – original draft. MH-G: Writing – original draft. EH-V: Resources, Writing – original draft, Funding acquisition, Supervision, Conceptualization, Writing – review & editing.

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References

1. Grieves M. Digital twin: manufacturing excellence through virtual factory replication. *White paper*. (2014) 1:1–7. Available online at: https://www.3ds.com/fileadmin/PRODUCTS-SERVICES/DELMIA/PDF/Whitepaper/DELMIA-APRISO-Digital-Twin-Whitepaper.pdf

2. Grieves M, Vickers J. Origins of the digital twin concept. *Fla Inst Technol*. (2016) 8:3–20.

3. Tao F, Qi Q. Make more digital twins. *Nature*. (2019) 573:490–1. doi: 10.1038/d41586-019-02849-1

4. Grieves MW. Digital twins: past, present, and future. In: Crespi N, Drobot AT, Minerva R, editors. *The Digital Twin*. Cham: Springer (2023). p. 97–121. doi: 10.1007/978-3-031-21343-4_4

5. Grieves M, Vickers J. Digital twin: mitigating unpredictable, undesirable emergent behavior in complex systems. In: Kahlen J, Flumerfelt S, Alves A, editors. *Transdisciplinary Perspectives on Complex Systems: New Findings and Approaches*. Cham: Springer (2016). p. 85–113. doi: 10.1007/978-3-319-38756-7_4

6. Liu W, Wu M, Wan G, Xu M. Digital twin of space environment: development, challenges, applications, and future outlook. *Remote Sens*. (2024) 16:3023. doi: 10.3390/rs16163023

7. Nikolas Noel GE. *Launches the Next Evolution of Wind Energy Making Renewables More Efficient, Economic: the Digital Wind Farm*. (2015). Available online at: https://www.ge.com/news/press-releases/ge-launches-next-evolution-wind-energy-making-renewables-more-efficient-economic (Accessed August 18, 2025).

8. Chevron Corporation. *The Good Twin: How Digital Doppelgängers are Driving Progress*. (2023). Available online at: https://www.chevron.com/newsroom/2023/q1/how-digital-doppelgangers-are-driving-progress (Accessed August 18, 2025).

9. Tom White. *Newcastle's "Digital Twin" to help City Plan for Disasters*. (2018). Available online at: https://www.theguardian.com/cities/2018/dec/30/newcastles-digital-twin-to-help-city-plan-for-disasters (Accessed August 18, 2025).

10. Observatory of Public Sector Innovation. *Virtual Singapore - Singapore's Virtual Twin*. (2018). Available online at: https://oecd-opsi.org/innovations/virtual-twin-singapore/ (Accessed August 18, 2025).

11. Talasila P, Gomes C, Mikkelsen PH, Arboleda SG, Kamburjan E, Larsen PG. Digital twin as a service (DTaaS): a platform for digital twin developers and users. In: 2023 *IEEE Smart World Congress (SWC)*. Portsmouth: IEEE (2023). p. 1–8. doi: 10.1109/SWC57546.2023.10448890

12. Talasila P, Gomes C, Vosteen LB, Iven H, Leucker M, Gil S, et al. Composable digital twins on digital twin as a service platform. *Simulation*. (2025) 101:287–311. doi: 10.1177/00375497241298653

13. Lehner D, Pfeiffer J, Tinsel EF, Strljic MM, Sint S, Vierhauser M, et al. Digital twin platforms: requirements, capabilities, and future prospects. *IEEE Softw*. (2021) 39:53–61. doi: 10.1109/MS.2021.3133795

14. Gil S, Mikkelsen PH, Gomes C, Larsen PG. Survey on open-source digital twin frameworks-a case study approach. *Softw: Pract Exp*. (2024) 54:929–60. doi: 10.1002/spe.3305

15. Laubenbacher R, Mehrad B, Shmulevich I, Trayanova N. Digital twins in medicine. *Nat Comput Sci*. (2024) 4:184–91. doi: 10.1038/s43588-024-00607-6

16. National Academies of Sciences, Engineering and Medicine. *Opportunities and Challenges for Digital Twins in Biomedical Research*. Washington, DC: National Academies Press (US). (2023).

17. Niarakis A, Laubenbacher R, An G, Ilan Y, Fisher J, Flobak Å, et al. Immune digital twins for complex human pathologies: applications, limitations, and challenges. *NPJ Syst Biol Appl*. (2024) 10:141. doi: 10.1038/s41540-024-00450-5

18. Björnsson B, Borrebaeck C, Elander N, Gasslander T, Gawel DR, Gustafsson M, et al. Digital twins to personalize medicine. *Genome Med*. (2019) 12:4. doi: 10.1186/s13073-019-0701-3

19. Laubenbacher R, Adler F, An G, Castiglione F, Eubank S, Fonseca LL, et al. Toward mechanistic medical digital twins: some use cases in immunology. *Front Digit Health*. (2024) 6:1349595. doi: 10.3389/fdgth.2024.1349595

20. Joslyn LR, Linderman JJ, Kirschner DE. A virtual host model of Mycobacterium tuberculosis infection identifies early immune events as predictive of infection outcomes. *J Theor Biol*. (2022) 539:111042. doi: 10.1016/j.jtbi.2022.111042

21. Jenner AL, Aogo RA, Alfonso S, Crowe V, Deng X, Smith AP, et al. COVID-19 virtual patient cohort suggests immune mechanisms driving disease outcomes. *PLoS Pathog*. (2021) 17:e1009753. doi: 10.1371/journal.ppat.1009753

22. Castiglione F, Celada F. *Immune System Modelling and Simulation*. Boca Raton, FL: CRC Press (2015). doi: 10.1201/b18274

23. Castiglione F, Bernaschi M. C-IMMSIM: playing with the immune response. In: *Proceedings of the Sixteenth International Symposium on Mathematical Theory of Networks and Systems (MTNS2004)*. Leuven: Katholieke Universiteit Leuven Belgium (2004). p. 1–7.

24. Ribeiro HA, Vieira LS, Scindia Y, Adhikari B, Wheeler M, Knapp A, et al. Multi-scale mechanistic modelling of the host defence in invasive aspergillosis reveals leucocyte activation and iron acquisition as drivers of infection outcome. *J R Soc Interface*. (2022) 19:20210806. doi: 10.1098/rsif.2021.0806

25. Jarrett AM, Kazerouni AS, Wu C, Virostko J, Sorace AG, DiCarlo JC, et al. Quantitative magnetic resonance imaging and tumor forecasting of breast cancer patients in the community setting. *Nat Protoc*. (2021) 16:5309–38. doi: 10.1038/s41596-021-00617-y

26. Wright L, Davidson S. How to tell the difference between a model and a digital twin. *Adv Mod Simul Eng Sci*. (2020) 7:13. doi: 10.1186/s40323-020-00147-4

27. Malik OP, Hope G, Cheng S. Some issues on the practical use of recursive least squares identification in self-tuning control. *Int J Control*. (1991) 53:1021–33. doi: 10.1080/00207179108953663

28. Chen CT. *Linear System Theory and Design*. Philadelphia, PA: Saunders College Publishing. (1984).

29. Kalman RE. *Contributions to the Theory of Optimal Control, Vol. 5*. Mexico City: Boletín de la Sociedad Matemática Mexicana (1960). p. 102–19.

30. Simon D. *Optimal State Estimation: Kalman, H Infinity, and Nonlinear Approaches*. Hoboken, NJ: John Wiley & Sons (2006). doi: 10.1002/0470045345

31. Besancon G, de León-Morales J, Huerta-Guevara O. On adaptive observers for state affine systems. *Int J Control*. (2006) 79:581–91. doi: 10.1080/00207170600552766

32. Dochain D. State and parameter estimation in chemical and biochemical processes: a tutorial. *J Process Control*. (2003) 13:801–18. doi: 10.1016/S0959-1524(03)00026-X

33. Kreisselmeier G. Adaptive observers with exponential rate of convergence. *IEEE Trans Automat Control*. (1974) 19:841. doi: 10.1109/TAC.1977.1101401

34. Afri C, Andrieu V, Bako L, Dufour P. State and parameter estimation: a nonlinear Luenberger observer approach. *IEEE Trans Automat Contr*. (2016) 62:973–80. doi: 10.1109/TAC.2016.2566804

35. Deng F, Chen J, Chen C. Adaptive unscented Kalman filter for parameter and state estimation of nonlinear high-speed objects. *J Syst Eng Electron*. (2013) 24:655–65. doi: 10.1109/JSEE.2013.00076

36. Zhang Q. Adaptive observer for multiple-input-multiple-output (MIMO) linear time-varying systems. *IEEE Trans Automat Contr*. (2002) 47:525–9. doi: 10.1109/9.989154

37. Luders G, Narendra K. An adaptive observer and identifier for a linear system. *IEEE Trans Automat Contr*. (1973) 18:496–9. doi: 10.1109/TAC.1973.1100369

38. Bastin G, Gevers MR. Stable adaptive observers for nonlinear time-varying systems. *IEEE Trans Automat Contr*. (1988) 33:650–8. doi: 10.1109/9.1273

39. Van Der Merwe R, Wan EA. The square-root unscented Kalman filter for state and parameter-estimation. In: *2001 IEEE International Conference on Acoustics, SPEECH, and Signal Processing. Proceedings (Cat. No. 01CH37221), Vol. 6*. Salt Lake City, UT: IEEE (2001). p. 3461–4. doi: 10.1109/ICASSP.2001.940586

40. Afshari HH, Gadsden SA, Habibi S. Gaussian filters for parameter and state estimation: a general review of theory and recent trends. *Signal Process*. (2017) 135:218–38. doi: 10.1016/j.sigpro.2017.01.001

41. Rodríguez A, Quiroz G, Femat R, Méndez-Acosta HO, de León J. An adaptive observer for operation monitoring of anaerobic digestion wastewater treatment. *Chem Eng J*. (2015) 269:186–93. doi: 10.1016/j.cej.2015.01.038

42. Basin M, Shi P, Calderon-Alvarez D. Optimal state filtering and parameter identification for linear stochastic time-delay systems. *IFAC Proc Vol*. (2008) 41:10246–51. doi: 10.3182/20080706-5-KR-1001.01734

43. Yaakov BS. Optimal simultaneous state estimation and parameter identification in linear discrete-time systems. *IEEE Trans Automat Contr*. (1972) 17:308–19. doi: 10.1109/TAC.1972.1100005

44. Togneri R, Deng L. Joint State and parameter estimation for a target-directed nonlinear dynamic system model. *IEEE Trans Signal Process*. (2003) 51:3061–70. doi: 10.1109/TSP.2003.819013

45. Nguyen NT, Nguyen TNT, Tong HN, Truong HVA, Tran DT. Dynamic parameter Identification based on the Least Squares method for a 6-DOF Manipulator. In: *Proceedings of 2023 International Conference on System Science and Engineering, ICSSE 2023*. (2023). p. 301–5. doi: 10.1109/ICSSE58758.2023.10227164

46. Sun Z, Yang Z. Application of joint parameter identification and state estimation to a fault-tolerant robot system. In: *IECON Proceedings (Industrial Electronics Conference)*. Melbourne, VIC: IEEE (2011). p. 498–503. doi: 10.1109/IECON.2011.6119361

47. Laubenbacher R, Niarakis A, Helikar T, An G, Shapiro B, Malik-Sheriff RS, et al. Building digital twins of the human immune system: toward a roadmap. *npj Digit Med*. (2022) 12:5. doi: 10.1038/s41746-022-00610-z

48. Wang H, Arulraj T, Ippolito A, Popel AS. From virtual patients to digital twins in immuno-oncology: lessons learned from mechanistic quantitative systems pharmacology modeling. *NPJ Digit Med*. (2024) 7:189. doi: 10.1038/s41746-024-01188-4

49. Vodovotz Y. Towards systems immunology of critical illness at scale: from single cell 'omics to digital twins. *Trends Immunol*. (2023) 44:345–55. doi: 10.1016/j.it.2023.03.004

50. Paul W. The immune system-complexity exemplified. *Math Model Nat Phenom*. (2012) 7:4–6. doi: 10.1051/mmnp/20127502

51. Levine S, Battisti T, Butz B, D'Souza K, Costabal FS, Peirlinck M. Dassault systèmes' living heart project. In: *Modelling Congenital Heart Disease: Engineering a Patient-specific Therapy*. Cham: Springer (2022). p. 245–59. doi: 10.1007/978-3-030-88892-3_25

52. Corral-Acero J, Margara F, Marciniak M, Rodero C, Loncaric F, Feng Y, et al. The 'Digital Twin' to enable the vision of precision cardiology. *Eur Heart J*. (2020) 41:4556–64. doi: 10.1093/eurheartj/ehaa159

53. Breton MD, Kanapka LG, Beck RW, Ekhlaspour L, Forlenza GP, Cengiz E, et al. A randomized trial of closed-loop control in children with type 1 diabetes. *N Engl J Med*. (2020) 383:836–45. doi: 10.1056/NEJMoa2004736

54. Man CD, Micheletto F, Lv D, Breton M, Kovatchev B, Cobelli C. The UVA/PADOVA type 1 diabetes simulator: new features. *J Diabetes Sci Technol*. (2014) 8:26–34. doi: 10.1177/1932296813514502

55. Suen Sc, Goldhaber-Fiebert JD, Brandeau ML. Risk stratification in compartmental epidemic models: where to draw the line? *J Theor Biol*. (2017) 428:1–17. doi: 10.1016/j.jtbi.2017.06.004

56. Wu H, Kumar A, Miao H, Holden-Wiltse J, Mosmann TR, Livingstone AM, et al. Modeling of influenza-specific CD8+ T cells during the primary response indicates that the spleen is a major source of effectors. *J Immunol*. (2011) 187:4474–82. doi: 10.4049/jimmunol.1101443

57. Miao H, Xia X, Perelson AS, Wu H. On identifiability of nonlinear ODE models and applications in viral dynamics. *SIAM Rev*. (2011) 53:3–39. doi: 10.1137/090757009

58. Alanis AY, Hernandez-Gonzalez M, Hernandez-Vargas EA. Observers for biological systems. *Appl Soft Comput*. (2014) 24:1175–82. doi: 10.1016/j.asoc.2013.10.028

59. Jhutty SS, Boehme JD, Jeron A, Volckmar J, Schultz K, Schreiber J, et al. Predicting influenza A virus infection in the lung from hematological data with machine learning. *Msystems*. (2022) 7:e00459–22. doi: 10.1128/msystems.00459-22

60. Hernandez-Vargas EA. Modeling kick-kill strategies toward HIV cure. *Front Immunol*. (2017) 8:995. doi: 10.3389/fimmu.2017.00995

61. Hernandez-Vargas EA, Middleton RH. Modeling the three stages in HIV infection. *J Theor Biol*. (2013) 320:33–40. doi: 10.1016/j.jtbi.2012.11.028

62. Hernandez-Vargas EA, Colaneri P, Middleton RH. Switching strategies to mitigate HIV mutation. *IEEE Trans Control Syst Technol*. (2013) 22:1623–8. doi: 10.1109/TCST.2013.2280920

63. Boianelli A, Nguyen VK, Ebensen T, Schulze K, Wilk E, Sharma N, et al. Modeling influenza virus infection: a roadmap for influenza research. *Viruses*. (2015) 7:5274–304. doi: 10.3390/v7102875

64. Hernandez-Vargas AE, Meyer-Hermann M. Innate immune system dynamics to influenza virus. *IFAC Proc Vol*. (2012) 45:260–5. doi: 10.3182/20120829-3-HU-2029.00029

65. Hernandez-Vargas EA, Wilk E, Canini L, Toapanta FR, Binder SC, Uvarovskii A, et al. Effects of aging on influenza virus infection dynamics. *J Virol*. (2014) 88:4123–31. doi: 10.1128/JVI.03644-13

66. Parra-Rojas C, Nguyen VK, Hernandez-Mejia G, Hernandez-Vargas EA. Neuraminidase inhibitors in influenza treatment and prevention-is it time to call it a day? *Viruses*. (2018) 10:454. doi: 10.3390/v10090454

67. Levy SB, Marshall B. Antibacterial resistance worldwide: causes, challenges and responses. *Nat Med*. (2004) 10:S122–9. doi: 10.1038/nm1145

68. Murray CJ, Ikuta KS, Sharara F, Swetschinski L, Aguilar GR, Gray A, et al. Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis. *Lancet*. (2022) 399:629–55. doi: 10.1016/S0140-6736(21)02724-0

69. Tetteh JN, Olaru S, Crauel H, Hernandez-Vargas EA. Scheduling collateral sensitivity-based cycling therapies toward eradication of drug-resistant infections. *Int J Robust Nonlinear Control*. (2023) 33:4824–42. doi: 10.1002/rnc.6528

70. Imamovic L, Sommer MO. Use of collateral sensitivity networks to design drug cycling protocols that avoid resistance development. *Sci Transl Med*. (2013) 5:204ra132. doi: 10.1126/scitranslmed.3006609

71. Anderson A, Kinahan MW, Gonzalez AH, Udekwu K, Hernandez-Vargas EA. Invariant set theory for predicting potential failure of antibiotic cycling. *Infect Dis Model*. (2025) 10:897–908. doi: 10.1016/j.idm.2025.04.001

72. Alavez-Ramírez J, Castellanos JRA, Esteva L, Flores JA, Fuentes-Allen JL, García-Ramos G, et al. Within-host population dynamics of antibiotic-resistant *M. tuberculosis*. *Math Med Biol*. (2007) 24:35–56. doi: 10.1093/imammb/dql026

73. Bhagunde PR, Nikolaou M, Tam VH. Modeling heterogeneous bacterial populations exposed to antibiotics: the logistic-dynamics case. *AIChE J*. (2015) 61:2385–93. doi: 10.1002/aic.14882

74. McCullers JA. The co-pathogenesis of influenza viruses with bacteria in the lung. *Nat Rev Microbiol*. (2014) 12:252–62. doi: 10.1038/nrmicro3231

75. Ballinger MN, Standiford TJ. Postinfluenza bacterial pneumonia: host defenses gone awry. *J Interferon Cytokine Res*. (2010) 30:643–52. doi: 10.1089/jir.2010.0049

76. Gordon S, Martinez FO. Alternative activation of macrophages: mechanism and functions. *Immunity*. (2010) 32:593–604. doi: 10.1016/j.immuni.2010.05.007

77. Mills CD, Ley K. M1 and M2 macrophages: the chicken and the egg of immunity. *J Innate Immun*. (2014) 6:716–26. doi: 10.1159/000364945

78. Duvigneau S, Sharma-Chawla N, Boianelli A, Stegemann-Koniszewski S, Nguyen VK, Bruder D, et al. Hierarchical effects of pro-inflammatory cytokines on the post-influenza susceptibility to pneumococcal coinfection. *Sci Rep*. (2016) 6:37045. doi: 10.1038/srep37045

79. Sharma-Chawla N, Stegemann-Koniszewski S, Christen H, Boehme JD, Kershaw O, Schreiber J, et al. *In vivo* neutralization of pro-inflammatory cytokines during secondary streptococcus pneumoniae infection post influenza a virus infection. *Front Immunol*. (2019) 10:1864. doi: 10.3389/fimmu.2019.01864

80. Almocera AES, Hernandez-Mejia G, Parra-Rojas C, Hernandez-Vargas EA. The trichotomy of pneumococcal infection outcomes in the Host. *Commun Nonlin Sci Numer Simul*. (2019) 73:1–13. doi: 10.1016/j.cnsns.2019.01.025

81. Kovatchev B, Cheng P, Anderson SM, Pinsker JE, Boscari F, Buckingham BA, et al. Feasibility of long-term closed-loop control: a multicenter 6-month trial of 24/7 automated insulin delivery. *Diabetes Technol Ther*. (2017) 19:18–24. doi: 10.1089/dia.2016.0333

82. Waight MC, Prakosa A, Li AC, Bunce N, Marciniak A, Trayanova NA, et al. Personalized heart digital twins detect substrate abnormalities in scar-dependent ventricular tachycardia. *Circulation*. (2025) 151:521–33. doi: 10.1161/CIRCULATIONAHA.124.070526

83. Coorey G, Figtree GA, Fletcher DF, Snelson VJ, Vernon ST, Winlaw D, et al. The health digital twin to tackle cardiovascular disease—a review of an emerging interdisciplinary field. *NPJ Digit Med*. (2022) 5:126. doi: 10.1038/s41746-022-00640-7

84. Gilbert AD, Kuo FY, Nuyens D, Pash G, Sloan IH, Willcox KE. Quasi-Monte Carlo methods for uncertainty quantification of tumor growth modeled by a parametric semi-linear parabolic reaction-diffusion equation. *arXiv [preprint]*. (2025) arXiv:2509.25753. doi: 10.48550/arXiv:2509.25753

85. Afri C, Andrieu V, Bako L, Dufour P. Identification of linear systems with nonlinear luenberger observers. In: *2015 American Control Conference (ACC)*. Chicago, IL: IEEE (2015). p. 3373–8. doi: 10.1109/ACC.2015.7171853

86. Evensen G. The ensemble Kalman filter for combined state and parameter estimation. *IEEE Control Syst Mag*. (2009) 29:83–104. doi: 10.1109/MCS.2009.932223

87. Perrier M, De Azevedo SF, Ferreira EC, Dochain D. Tuning of observer-based estimators: theory and application to the on-line estimation of kinetic parameters. *Control Eng Pract*. (2000) 8:377–88. doi: 10.1016/S0967-0661(99)00164-1

88. Olay-Blanco A, Rodriguez-Liñan A, Quiroz G. Parameter and State Estimation of a Mathematical Model of Carbohydrate Intake. In: *IFAC-PapersOnLine. 2nd IFAC Conference on Modelling, Identification and Control of Nonlinear Systems MICNON 2018, Vol. 51*. (2018). p. 73–8. doi: 10.1016/j.ifacol.2018.07.257